

Interactive Cues on Geometry Learning in a Virtual Reality Environment for K-12 Education

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ABSTRACT

Spatial imagery and reasoning are challenging for K-12 students studying geometry. Virtual reality (VR) technology, known for its high level of interactivity and immersion, has increasingly been used in research to facilitate geometry learning. However, most previous VR environments for geometry learning lack dynamic cues to aid in the observation, imagination and spatial reasoning of students. In order to address this issue, we propose a demo of three interactive cues for students to successfully complete tasks in geometry learning: dynamic arrow, painting, and projecting cues. When students interact with 3D objects, dynamic arrow cues indicate the direction of rotation of the object. Additionally, students can employ painting cues to differentiate the faces of objects. Projecting cues allow for the projection of objects onto a plane. The purpose of these visual cues is to direct students' attention, facilitate mental imagery and reduce cognitive load in geometry learning.

Keywords: Geometry, Cues, Virtual Reality

Index Terms: Human-centered computing—Virtual Reality

1 INTRODUCTION

Spatial ability is regarded as one of the fundamental human abilities. Empirical research has consistently demonstrated that the development of spatial skills can lead to improved performance in STEM [1]. The tasks associated with geometry require spatial abilities, such as spatial imagery and reasoning skills, from the students. However, these skills can be challenging for K-12 students. Virtual reality (VR) technology, known for its high level of interactivity and immersion, has been increasingly used in research aimed at enhancing spatial abilities and geometric learning [2–7]. However, most previous VR environments for learning geometry lacked dynamic cues to aid students in observation, imagination, and spatial reasoning. Geometry learning tasks in VR systems are often complex for K-12 students in early grades. In addition, they may become distracted by unimportant elements of the VR environment and neglect essential components of the learning task.

In a meta-analysis conducted by David et al. [8], the impact of incorporating cues into multimedia learning materials was examined. The study analyzed 29 experimental studies involving a total of 2726 participants. The results indicated that the inclusion of cues in multimedia learning materials had a positive effect on learning outcomes. A study of implementing cues in multimedia learning indicated that visual cues can direct attention to the correct areas of the screen at the right time, thereby promoting learning [9]. In particular, when learning visually cued graphics, students learned more efficiently than their peers in uncued-static condition [10].

However, it should be noted that research on the utilization of cueing within the VR context is still in its early stage.

To address the issue of lack of VR cuing in assisting geometry learning for K-12 students, we design three interactive cues in a desktop application. These cues, namely dynamic arrow, painting, and projecting cues, are intended to enhance students' ability to successfully complete tasks related to geometry learning. Specifically, dynamic arrow cues are utilized to indicate the direction of rotation of 3D objects, painting cues are employed to differentiate the various faces of objects, and projecting cues enable the projection of objects onto a plane. The ultimate goal of these visual cues is to direct students' attention, promote mental imagery, and mitigate cognitive load associated with geometry learning.

2 GEOMETRY LEARNING DESKTOP VR APPLICATION AND DYNAMIC CUES

A desktop virtual reality (desktop VR) application was developed for the purpose of facilitating geometry learning among students aged 7-8. The design of the VR materials was based on the curriculum outlined in the instructor's textbook and lecture slides. The primary objective of the application is to provide instruction on the identification and manipulation of geometric solids and shapes. The application was developed using Unity [11] and 3ds Max [12], and all scenes and objects within the application were created using these software programs.

In this system, we propose three interactive dynamic cues to assist students in observing and interacting with geometric objects, including dynamic arrow, painting, and projecting cues.

- **Dynamic arrow cues** The acquisition of spatial understanding is a crucial aspect of the cognitive development of 7-8-year-old students. They need to observe the solid from 6 sides and be able to imagine the plane obtained from looking at the solid from different directions. To facilitate this process, we design dynamic arrow cues that allow students to perceive the rotational direction of a solid object when viewed from different perspectives. Striking three-dimensional dynamic arrow indicate the object's four rotational directions, as shown in Figure 1 (top left). The students can rotate the object by clicking on the vertical or horizontal rotation buttons, thereby allowing them to actively engage in building their understanding of the rotation of geometric objects.
- **Painting cues** In order to enhance the students' ability to distinguish different faces or shapes of a geometric object, we have implemented a painting cue. Painting cues are used to differentiate various components of the object, as shown in Figure 1 (top middle). The students are able to select their preferred colors to fill in each part of the geometric object within the virtual reality (VR) system. The utilization of color cues serves to facilitate the identification of the object's sides when it is rotated to different angles, thereby promoting a more efficient understanding of the object's various perspectives.
- **Projecting cues** The acquisition of spatial understanding, particularly in regards to the three-view representation of geometric objects, can be a challenging task for K-12 students in the

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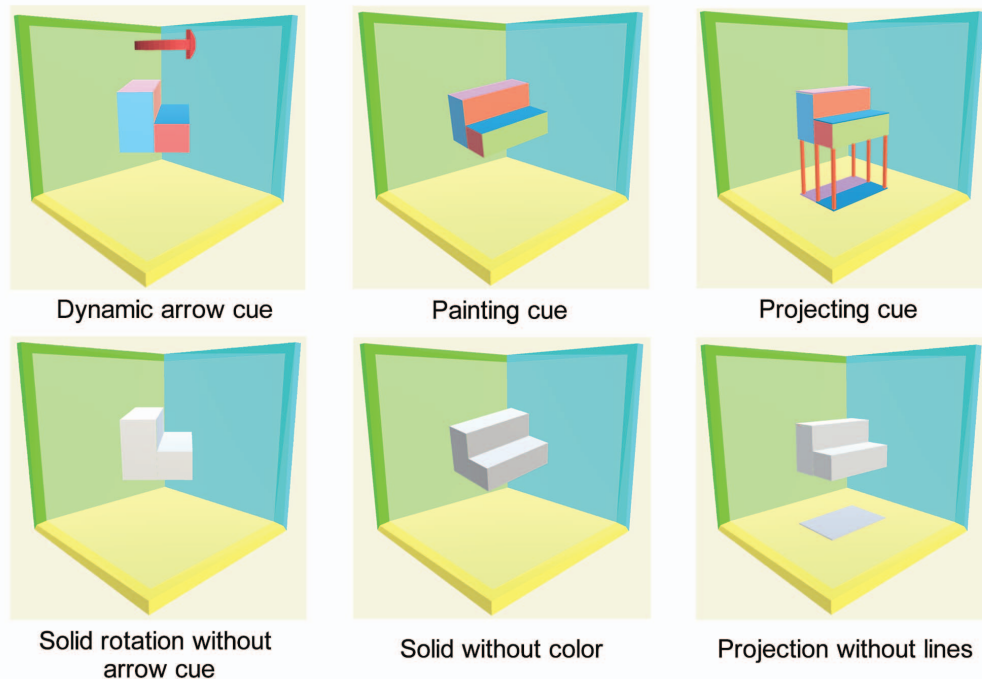


Figure 1: Three Dynamic Cues. Top left: dynamic arrow cue. Top middle: painting cue. Top Right: projecting cue. Bottom: the counterpart without cues.

early stage. To address this difficulty, we have implemented a projection cue within the system. This cue involves the extension of multiple vertical lines from each vertex of each view, which form two-dimensional figures on three baffles, as shown in Figure 1 (top right). This projecting cue serves to enhance the students' understanding of the relationship between solid objects and their corresponding planes. By observing the three-dimensional figures, students are able to easily construct their own mental representation of the geometric object, thereby promoting a deeper understanding of its properties.

3 CONCLUSION AND FUTURE DIRECTIONS

In this preliminary work, we propose a novel approach for enhancing the learning of geometry in a desktop virtual reality (VR) system through the implementation of three interactive dynamic cues, namely, dynamic arrow, painting, and projecting cues. The purpose of these cues is not only to direct the attention of students towards crucial aspects of the task, but also to augment the spatial information in VR, thereby facilitating the development of mental imagery and the ability to reason.

As part of our ongoing research, we aim to conduct an extensive examination of the effect of the proposed interactive dynamic cues (i.e. dynamic arrow, painting, and projecting cues) on the performance and spatial ability of K-12 students within a desktop virtual reality (VR) learning environment. To achieve this goal, we plan to employ a range of methodological approaches including the measurement of factors such as motivation, learning experience, attitudes towards learning, and cognitive load in order to gain a deeper understanding of students' perceptions of the VR system and cues. Our expectation is that when learning geometry with dynamic cues, students will have better spatial ability, geometry performance and more active learning attitude than their peers without cues. Additionally, we intend to employ camera-based eye-tracking technology to monitor the students' processing of the cues, in order to gain further

insight into the effectiveness of the proposed approach.

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